Concentric escutcheon pair comparison

CB01 | R005 reference geometry | millimeters | 2026-10-15





Pair	Main / diverter OD	Border widths	Gap	Worst main margin
C1	220 / 120	17.92 / 10.79	5.24	7.41
C2	214.15 / 128.425	15.00 / 15.00	3.95	4.48

Both pairs use circular outlines with zero intentional offset from their own covers. C1 keeps more main coverage. C2 gives both covers the same 15 mm border. Main cover OD is measured; diverter cover OD and installed centers remain provisional.

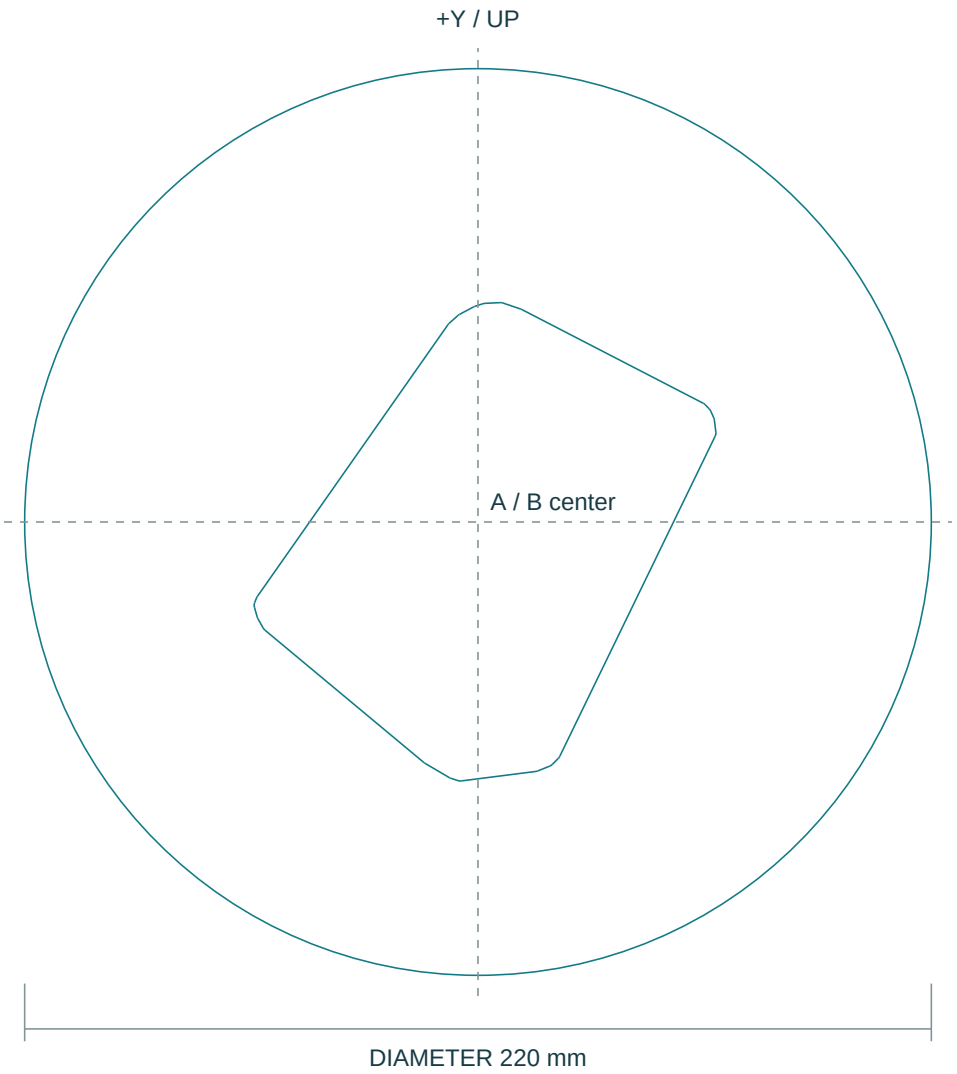
Margins are projected distances beyond traced damage, after existing trace allowances. They are not proven sound sealing land. Both options fall below the earlier exploratory 10 mm main-land target. Assembly images show a conditional appearance using official Kohler exterior meshes; actual stud/foam/depth fit is unresolved.

Sensitivity example: if each actual cover center were within 1 mm of its model location, worst-case gaps would be 3.24 mm (C1) and 1.95 mm (C2). That 1 mm bound has not been measured. A 2 mm bound would erase the C2 gap. Scan scale and support uncertainty remain additional.

MAIN-C1 | custom backing plate

CB01 | R005 reference geometry | millimeters | 2026-10-15

DATUM A: rear flat face, Z=0. DATUM B: outer-circle center, X=Y=0. +X right and +Y up when viewed from the shower. The visible outline is concentric with the factory cover.



Stock trial	Thickness (mm)	Use / material
SS060	1.524	316 stainless steel
SS125	3.175	316 stainless steel
PROTO3	3.0	Clear acrylic or printed PLA/PETG dry-fit prototype only

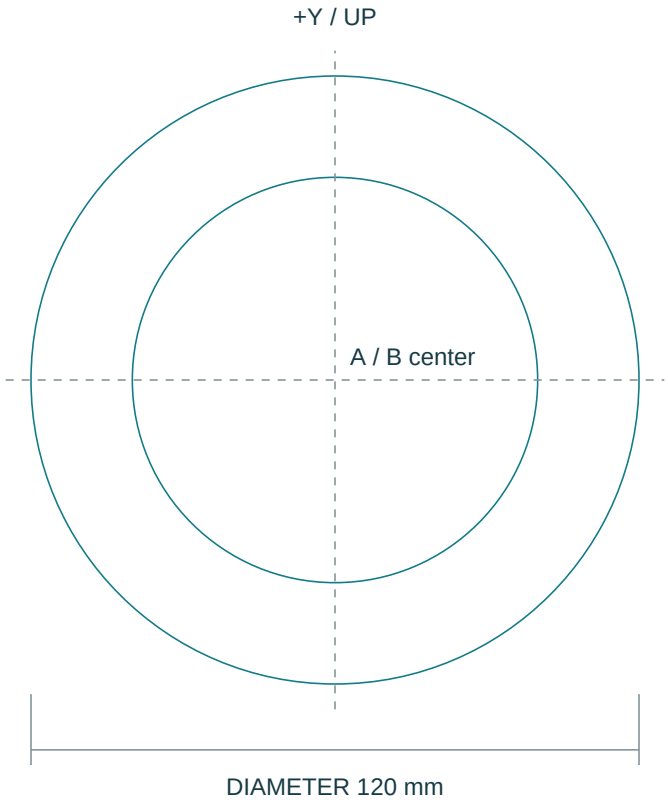
CUT GEOMETRY: STEP/DXF controls; do not scale this drawing. Two closed contours only. Main aperture uses the common coordinate schedule on sheet 6. No separate stud holes.

Finish concept: satin/brushed face, direction vertical; color match to Kohler BN is unverified. Deburr both faces; no sharp edges or burrs against foam/stone. Shop to confirm size, thickness, flatness and edge tolerances. No numerical manufacturing tolerance or seal compression has been released.

DIV-C1 | custom backing plate

CB01 | R005 reference geometry | millimeters | 2026-10-15

DATUM A: rear flat face, Z=0. DATUM B: outer-circle center, X=Y=0. +X right and +Y up when viewed from the shower. The visible outline is concentric with the factory cover.



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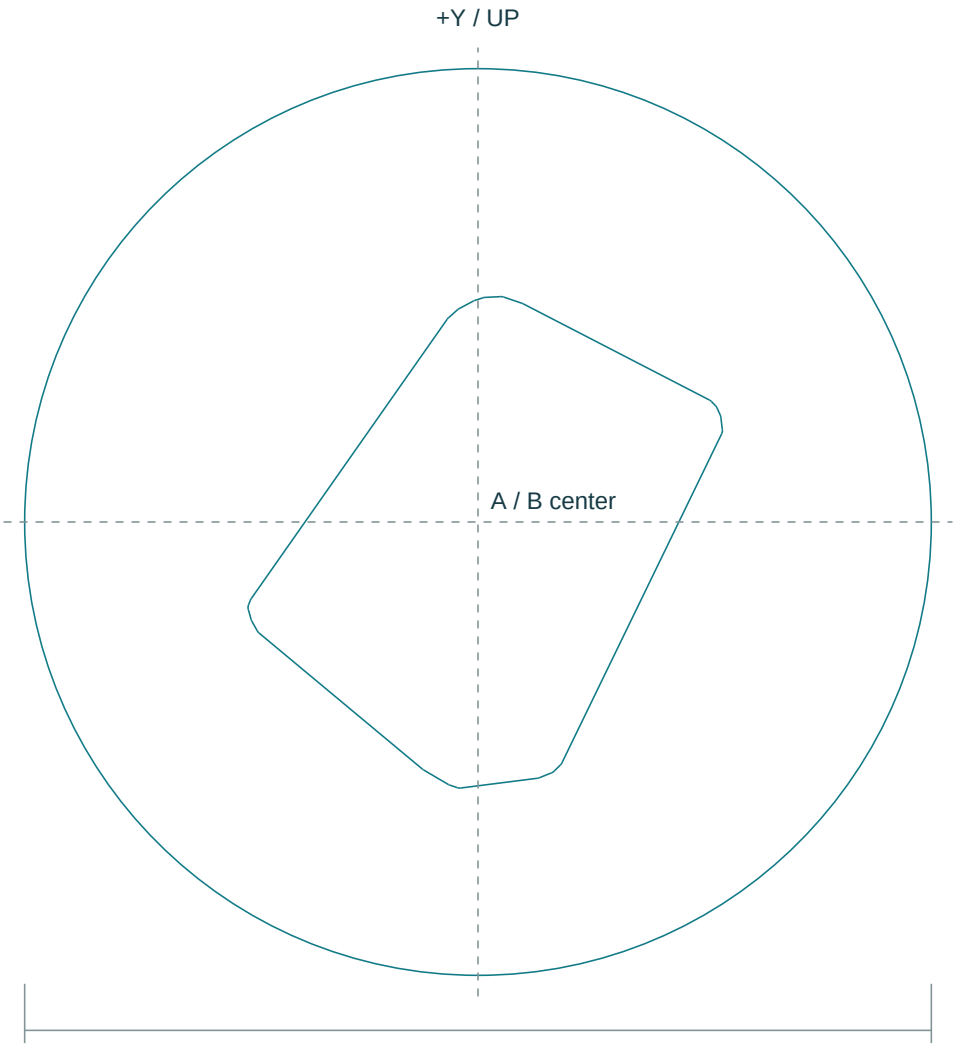
CUT GEOMETRY: STEP/DXF controls; do not scale this drawing. Two closed contours only. Central aperture is diameter 80 mm, concentric. This diverter clearance remains a trial.

Finish concept: satin/brushed face, direction vertical; color match to Kohler BN is unverified. Deburr both faces; no sharp edges or burrs against foam/stone. Shop to confirm size, thickness, flatness and edge tolerances. No numerical manufacturing tolerance or seal compression has been released.

MAIN-C2 | custom backing plate

CB01 | R005 reference geometry | millimeters | 2026-10-15

DATUM A: rear flat face, Z=0. DATUM B: outer-circle center, X=Y=0. +X right and +Y up when viewed from the shower. The visible outline is concentric with the factory cover.



Stock trial	Thickness (mm)	Use / material
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SS125	3.175	316 stainless steel
PROTO3	3.0	Clear acrylic or printed PLA/PETG dry-fit prototype only

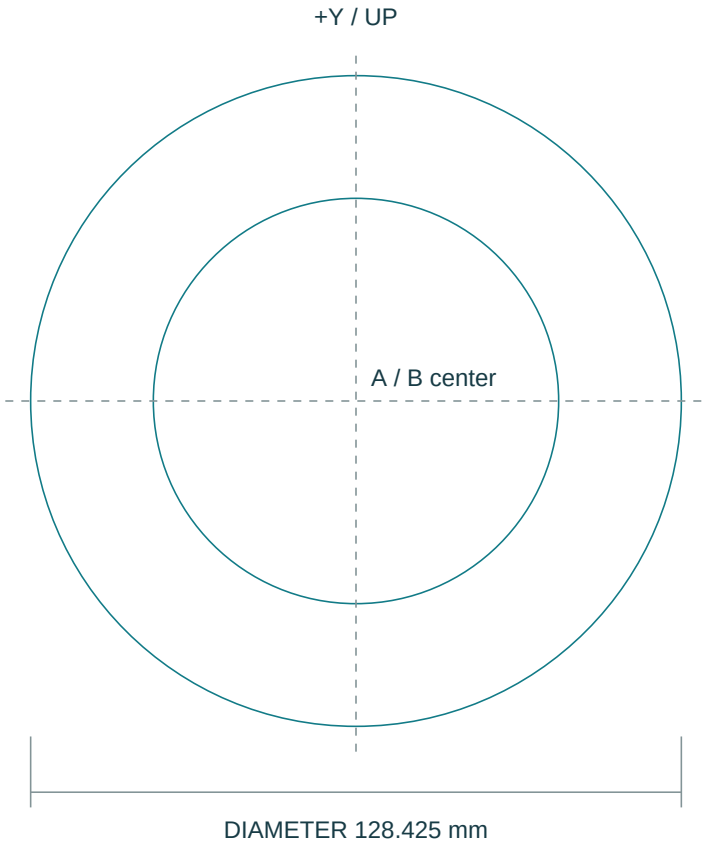
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DIV-C2 | custom backing plate

CB01 | R005 reference geometry | millimeters | 2026-10-15

DATUM A: rear flat face, Z=0. DATUM B: outer-circle center, X=Y=0. +X right and +Y up when viewed from the shower. The visible outline is concentric with the factory cover.



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Common main aperture | coordinate schedule

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The two main plates use the same insertion aperture in installed coordinates. Below are the base polygon vertices relative to each concentric cover center; apply an OUTWARD 1.75 mm round offset. The supplied STEP/DXF contains the final offset contour. Rounded table values are reference only; files control. Traverse P01-P20 and close.

Point	X (mm)	Y (mm)	Point	X (mm)	Y (mm)
P01	-4.306	-61.094	P11	1.764	51.304
P02	-6.110	-60.496	P12	5.473	51.483
P03	-11.992	-57.034	P13	9.833	50.002
P04	-50.592	-24.884	P14	53.888	27.240
P05	-51.893	-22.570	P15	54.866	26.214
P06	-52.565	-20.196	P16	55.621	24.542
P07	-52.123	-19.097	P17	55.979	21.595
P08	-5.837	46.897	P18	18.253	-56.117
P09	-3.663	48.835	P19	16.757	-57.609
P10	-0.250	50.649	P20	13.836	-58.787

Do not recenter this aperture within the custom outline independently. It is registered to captured hardware; the outer circle is registered to the inferred cover center. Later physical registration may change their relationship.

The unchanged F005 aperture cleared all 51,929 checked forward-surface triangle projections. That test concerns the captured mesh only, not hidden surfaces or actual scale. The rear foam band remains untraced; the aperture may cross the real seal land.

Drawing datums and cut path are not installation locators. Before release, use the real cover as a reference and verify equal exposure around its perimeter during a thickness-correct dry fit. If center offset is e, ideal circular border variation is 2e peak-to-peak. No allowable e has been selected.

Exploded order | unchanged hardware concerns

CB01 | R005 reference geometry | millimeters | 2026-10-15



From the wall outward

1. Qualified support / seal detail - unresolved.
2. Custom backing plate.
3. Existing factory foam and inner plate.
4. Factory nuts behind the cover.
5. Decorative cover, adapters and handles.

Exploded positions are display offsets, not installed dimensions. The pink foam is a footprint hypothesis. F005 had stud-region clashes with its reference plates; none at the controls in those tested positions. That conditional result does not establish handle engagement or water sealing.

The reported studs touching wood have not been traced to their rear ends. No cutting, drilling, valve relocation or hardware adjustment is selected. Separate accessible support screws are visible in the front photographs.

Covers and handles use official Kohler exterior CAD for appearance; their hidden rear interfaces are not measurement authority. Original scan/proxy geometry, topology, UVs and transforms were preserved in the saved Blender file.

Fit record needed before a metal release

CB01 | R005 reference geometry | millimeters | 2026-10-15

New caliper center readings are preserved as a separate reconciliation study. They have not silently moved the captured valve, studs, aperture or cover registration used in these files.

Tip-center pair	Reported mm
U to L	80
NE to SW	109.8
U to NE	51.4
L to NE	80.25
U to SW	81.14
L to SW	54.7

U/L = upper/lower control; NE/SW = upper-right/lower-left trim stud. The user confirms 3D end-tip-center distances at unequal depths. These are not wall-plane hole spacings. See sheet 11 for depth correction; contact repeatability remains unknown.

1. Rear foam: trace both edges on the full-size template; confirm continuous landing, condition and intended drainage.

2. Hardware: measure stud diameter and axes at the intended plate plane; measure stud/control projection and the actual cover cavity/nut seats.

3. Placement: check real cover centers, diverter OD, equal visible borders, worst gouges, inter-plate clearance and sound support.

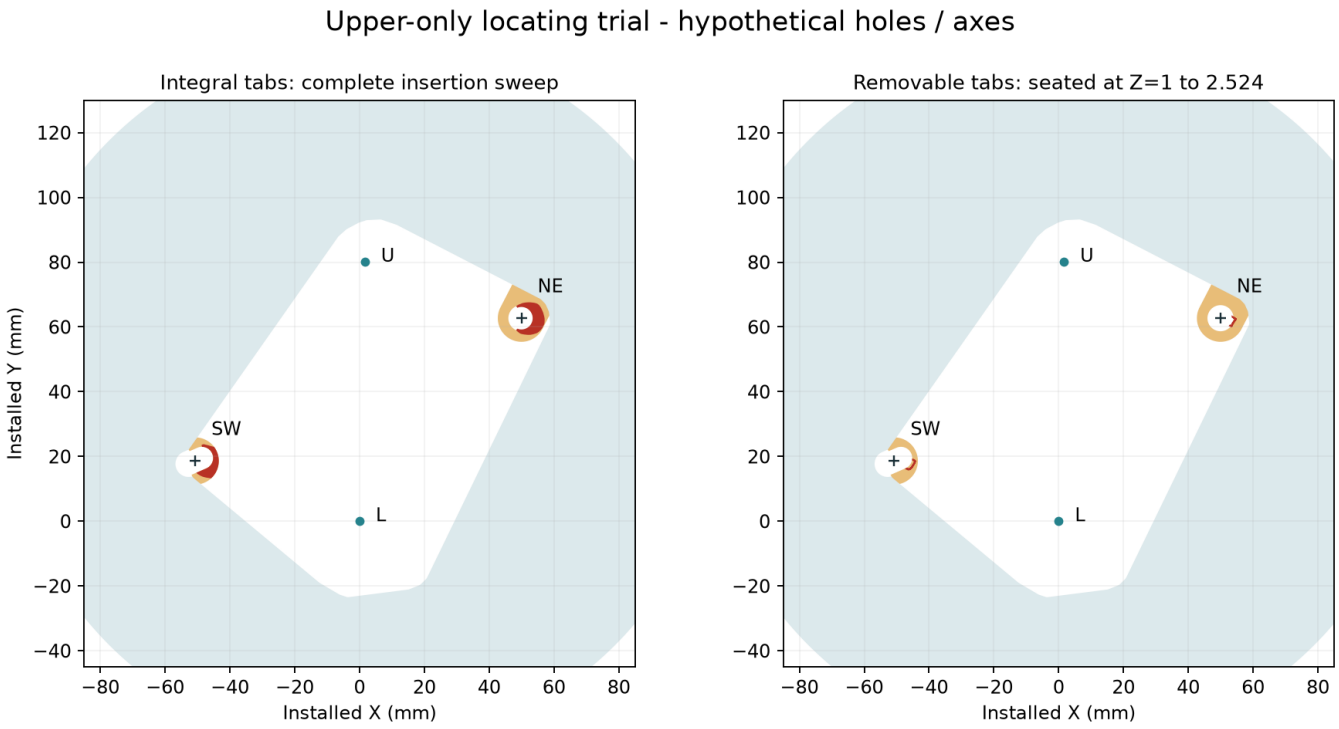
4. Thickness: PROTO3 is 1.476 mm thicker than SS060 and 0.175 mm thinner than SS125. A 3 mm prototype cannot prove the thin-metal stack; use the selected actual thickness for final dry fit.

5. Assembly: verify factory plate seating, foam compression, handle engagement/travel, drainage, removal and tool access.

6. Release: resolve the support/seal construction, material/finish and shop tolerances; obtain revision-specific approval. Then inspect the received parts and test for water entry after installation.

Upper-only locating tabs | diagnostic trial

CB01 | R005 reference geometry | millimeters | 2026-10-15



Amber: added tab metal. Red: projected captured-surface interference. Not a released mounting design.

A concrete C1 upper-plate trial retains metal at the two fitted stud positions: an exploratory 7.5 mm round hole at NE and a 7.5 x 12 mm slot at SW, along the stud-pair line, in 14 mm wide tabs. These are design trials, not measurements or released hole sizes. The lower companion stays unchanged.

Test against unchanged captured surfaces	Projected intersection
Integral-tab insertion sweep	54.86 mm2
Final slab Z=1 to 2.524 mm	0.42 mm2

The trial holes do not clear the unchanged scanned stud regions. This compares fitted tip locations with captured surfaces and assumed normal axes; it does not prove the real tab idea is impossible. The small seated conflict also means removable bridges do not yet have a proven fit. Actual stud diameter, inclination and rear/cover space are decisive.

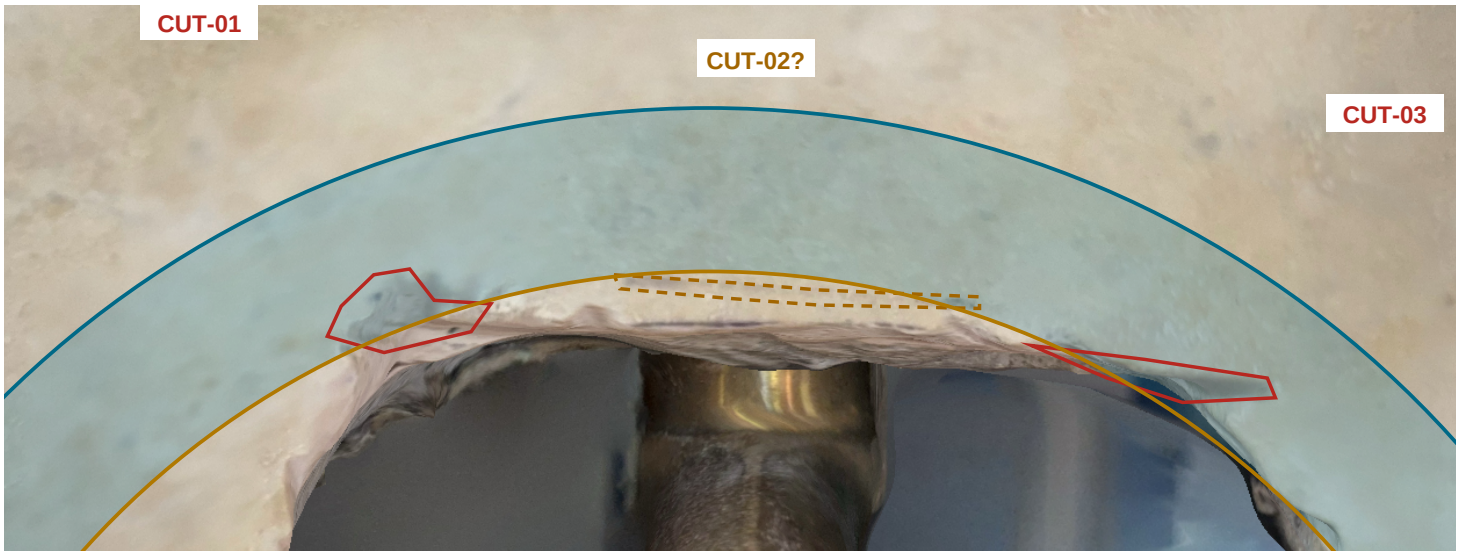
Through-hole capture is the current diagnostic approach. Threaded engagement is not selected: thread identity, engagement length and assembly method are unknown. Nut/washer bearing and restraint remain unverified. Clearance and a slot allow movement; they do not by themselves guarantee a concentric decorative cover.

The diagnostic STEP/STL is separate from all quote packages. No added attachment holes, stud cutting or factory-hole enlargement is selected. Compare the open-aperture template first, then reconcile the real stud axes before a locating prototype.

Top gouges beneath the visible border

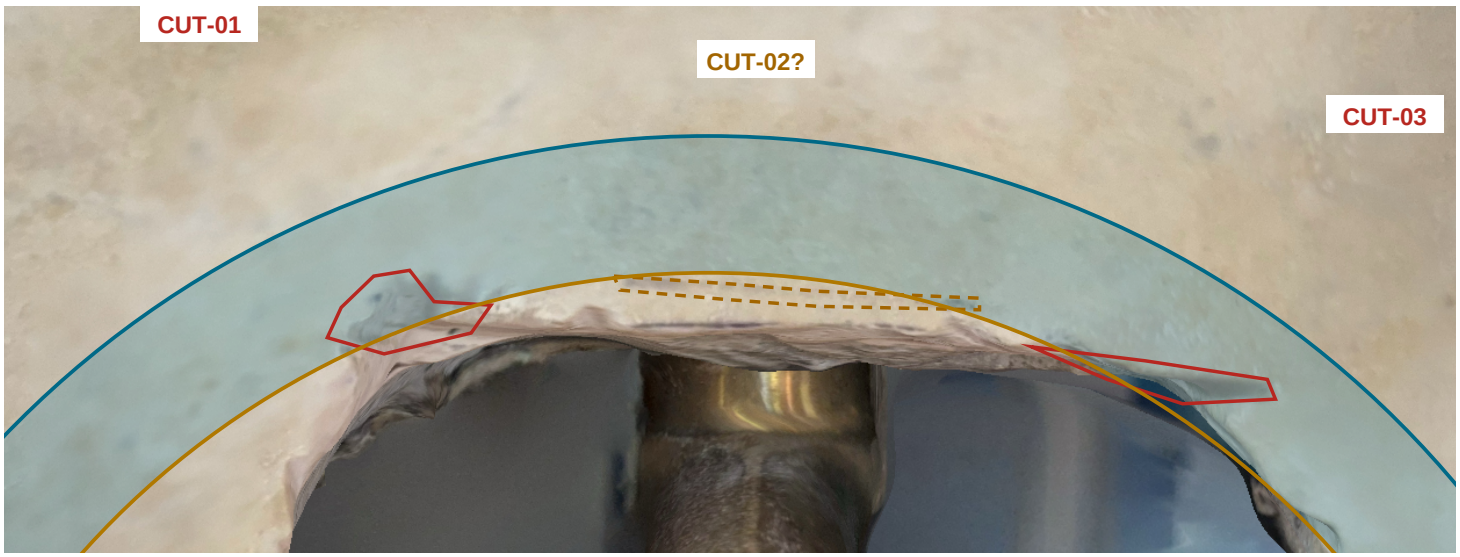
CB01 | R005 reference geometry | millimeters | 2026-10-15

C1 - top damage reveal



CUT-03: 9.02 mm beyond the factory-cover edge; 8.91 mm to the custom outer edge (7.41 mm after the 1.5 mm tracing allowance).

C2 - top damage reveal



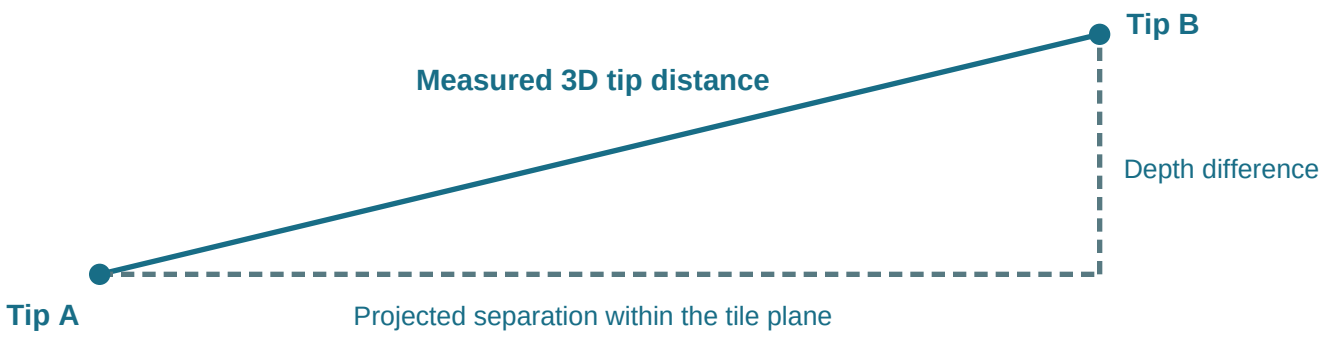
CUT-03: 9.02 mm beyond the factory-cover edge; 5.98 mm to the custom outer edge (4.48 mm after the 1.5 mm tracing allowance).

Blue: exposed custom border. Gold: factory-cover edge. Red: retained cut traces. Dashed amber CUT-02 remains uncertain. These are registered evidence overlays on the original scan render, not a finished transparent plate. The browser damage view also includes the full opening and the uncertain CUT-07 mark. Margins do not prove sound support or sealing.

Tip distances and the tile plane

CB01 | R005 reference geometry | millimeters | 2026-10-15

The six caliper readings are straight-line distances between the centers of projecting ends. Their different depths are part of each reading. Preserve these original numbers; compare them with 3D scan distances first.



d_XY = sqrt(d_3D squared - delta_Z squared) | schematic, not to scale

Pair	Raw 3D	Scan delta Z	Conditional XY	Reduction
U-L	80.00	1.447	79.987	0.013
NE-SW	109.80	4.065	109.725	0.075
U-NE	51.40	1.278	51.384	0.016
L-NE	80.25	2.725	80.204	0.046
U-SW	81.14	5.343	80.964	0.176
L-SW	54.70	6.790	54.277	0.423

All values above are mm. The converted values use unchanged baseline scan depths, so they are conditional illustrations, not accepted CAD dimensions. Even projected tip centers can differ from stud-axis locations at the plate plane when a stud is inclined.

Allowing depth to vary, all six reported lengths fit a 3D four-tip arrangement exactly. Both mirror branches are retained as separate diagnostic estimates. The distances alone do not locate that arrangement relative to the wall; matching the same six inputs does not establish measurement accuracy. The earlier fixed-depth fit retains scan depths and is only a sensitivity trial.

Keep four quantities separate: measurement/registration uncertainty, clearance needed to slide and seat the assembly, shop cutting/position tolerance, and the sealing/support land. Neither the depth correction nor a small fitting residual is a machining tolerance. No numerical tolerance has been released.